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*Article*

# Prediction and Factors Interpretation of Wildfire Susceptibility in Southern California Incorporating Remote Sensing Data into Machine Learning Approaches

**Mohammadsadegh Beheshti**

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**Abstract**

Accurate wildfire susceptibility prediction in Southern California is vital for regional hazard mitigation and emergency management. Despite the current availability of machine learning (ML) models for predicting wildfire susceptibility, there is a lack of high-performance modeling and interpretation over large amounts of historical fire data for this region. This study evaluates seven ML algorithms—Support Vector Classification (SVC), Logistic Regression (LR), Artificial Neural Network (ANN), K-Nearest Neighbors (KNN), Random Forest (RF), Gradient Boosting (GB), and Light Gradient Boosting Machine (LGBM)—using 18 geo-environmental and anthropogenic contributing factors in Riverside County from 2012 to 2023. Using a standard random train-test split, the top-performing LGBM model achieved an AUC of 0.97, an accuracy of 91.65%, and an F1-score of 91.51%. However, spatial evaluation via success and prediction rate curves revealed a more conservative and realistic generalizability (AUC = 0.748 and 0.740, respectively), underscoring the localized, non-portable nature of the regional predictors. Shapley Additive Explanations (SHAP) indicated that multi-year rainfall norms, solar radiation, distance to urban centers, and elevation were the primary drivers of model output. This framework provides a benchmark for regional susceptibility mapping while highlighting the essential constraints of spatial validation in fire engineering applications.

**Keywords:** machine learning; wildfire susceptibility; SHAP; wildfire prevention

## 1. Introduction

Wildfires burn approximately 420 million hectares (Mha) of global landscape annually, causing multi-billion dollar infrastructure damages and altering ecosystems [1]. Wildfire ignition and propagation are driven by a complex interplay of environmental factors, including topography, fuel conditions, weather paradigms, and spatial proximity to human infrastructure. Within wildland–urban interface (WUI) environments, wildfire susceptibility mapping serves as a critical cornerstone for broader risk assessment frameworks. Rather than establishing definitive causal links, these spatial susceptibility predictions identify areas with a high historical likelihood of fire occurrence, thereby mapping where human exposure intersects with volatile environmental conditions.

This approach complements recent performance-based WUI resilience networks. For instance, [2] established a holistic theoretical framework that leverages spatial infrastructure buffers to index community exposure and intervention accessibility during fire emergencies. Similarly, [3] demonstrated that fusing remote sensing-derived vegetation metrics with anthropogenic covariates captures the multi-driver dynamics of fire across complex natural-urban interfaces. Fusing

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geographic information systems (GIS) with predictive modeling is therefore imperative for targeting ecological vulnerability mitigation and localized suppression strategies.

Traditional approaches like the Analytic Hierarchy Process (AHP) or standard Logistic Regression (LR) often exhibit limited predictive accuracy due to their strict parametric assumptions and inability to capture non-linear feature interactions. Consequently, recent research has pivoted toward machine learning (ML) paradigms to enhance predictive benchmarks.

Previous studies demonstrate strong predictive capabilities across diverse international regions, though model accuracy remains highly sensitive to localized environmental baselines. Linear and non-linear algorithms—ranging from traditional Support Vector Machines (SVM), which encompass both Support Vector Classification (SVC) and Support Vector Regression (SVR), to deep learning (DL) architectures—have been widely tested. For example, [4] utilized a Random Forest (RF) algorithm with nine geo-environmental inputs to map fire susceptibility in Italy, noting that cross-validation provided a stable evaluation of model variance though it could not drive the structural Root Mean Squared Error (RMSE) below 0.50. Ensemble decision-tree frameworks and gradient-boosting methods have regularly yielded high random-split accuracies across variable sample sizes. In Southern France, [5] mapped ignition patterns using a 36 controlling factors RF model to achieve an F1-score of 0.87. Similar high-performance benchmarks (AUC ≥ 0.89) have been reported using tree-based and hybrid optimization models across diverse landscapes, such as in Amol [6], and the Similipal Tiger Reserve, [7].

Despite these high reported metrics (often exceeding 90% AUC under random test validation), a significant methodology gap remains in regional wildfire susceptibility mapping. Current frameworks rarely address high model performance, interpretability and diverse number of controlling features at the same time. To address that, this study evaluates and interprets seven distinct ML methods categorized across four algorithm families over a large historical record (2012–2023) and 18 diverse controlling factors as an example in Riverside County, Southern California:

1. Linear Models: Support Vector Classification (SVC) and Logistic Regression (LR).
2. Non-parametric / Non-linear Models: Artificial Neural Networks (ANN) and K-Nearest Neighbors (KNN).
3. Ensemble Bagged Trees: Random Forest (RF).
4. Gradient Boosting Trees: Gradient Boosting (GB) and Light Gradient Boosting Machine (LGBM).

The models are evaluated using standard pixel-based performance metrics paired with threshold-independent success and prediction rate curves to measure spatial generalizability. To resolve the black-box nature of the top-performing tree-based algorithms, the TreeSHAP formulation of the Shapley Additive Explanations framework [8] is used to unpack the operational importance and directional influence of each landscape predictor.

## 2. Materials and Methods

### *2.1. Study Area and Historical Wildfire Data*

The study area covers approximately 7,310 km² in Riverside County, Southern California, USA, as shown in Figure 1. The coordinates for the area range from 33.42612° to 34.03427° latitude and -117.67655° to -116.07585° longitude. The region's mountains are primarily located on the eastern side, with elevations spanning from -37 m to 3,283 m and temperatures ranging from 2 °F to 109 °F (-16.7 °C to 42.8 °C). Wildfire causes vary from lightning to human activities. This study, with the exception of Section 3.4.2, was conducted using historical wildfire data from the Visible Infrared Imaging Radiometer Suite (VIIRS), distributed by the Fire Information for Resource Management System (FIRMS) for the period of 2012–2023, as shown in Figure 1(b) (https://firms.modaps.eosdis.nasa.gov/). Annual VIIRS active fire data (2012–2023) were acquired from NASA FIRMS and clipped to the Riverside County boundary. Data were preprocessed to remove false alarms by retaining only nominal and high-confidence detections, eliminating duplicates, and applying spatial thinning to account for the 375m sensor resolution. These wildfire data consist of 6,366 points labeled as 1, while

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another 6,366 points randomly generated via ArcGIS in areas outside of known wildfires were labeled as 0. In total, 12,732 points were used with 0 representing no fire and 1 representing wildfire locations which were randomly divided into 70% training and 30% test sets for the training, testing, and comparison of the ML models.

Since VIIRS data may not include fires that were short-lived or did not generate sufficient heat, such as grass fires, the highest-performing model determined from the previous comparison was used to calculate wildfire susceptibility for a set of historical wildfire data larger than 2 acres and prescribed fires from 2012 to 2023, using the CAL FIRE Riverside Unit database. Figure 1(c) displays these data, including 1,800 points representing the centroids of the fire perimeters. Compared to the VIIRS data, the unit and prescribed fire centroids were more dispersed throughout the area due to the different acquisition methods for each dataset. These points were labeled as 1, and 1,800 random points were generated and labeled as 0 using ArcGIS in areas outside the recorded wildfire perimeters. To ensure reproducibility and capture spatial variability, non-fire points were generated using the ArcGIS Pro Create Random Points tool within the Riverside County study area (Figure 1). These samples were constrained by a 3-meter (9.8 feet) minimum separation to align with sub-resolution constraints on the 30-meter resampled grid and restricted to burnable land cover using an NDVI mask (> 0.2). To ensure reproducibility, non-fire point generation utilized a fixed random seed of 100. Consequently, a total of 3,600 points were divided into 70% training and 30% test sets to evaluate model behavior under distinct agency data streams in Section 3.4.2.

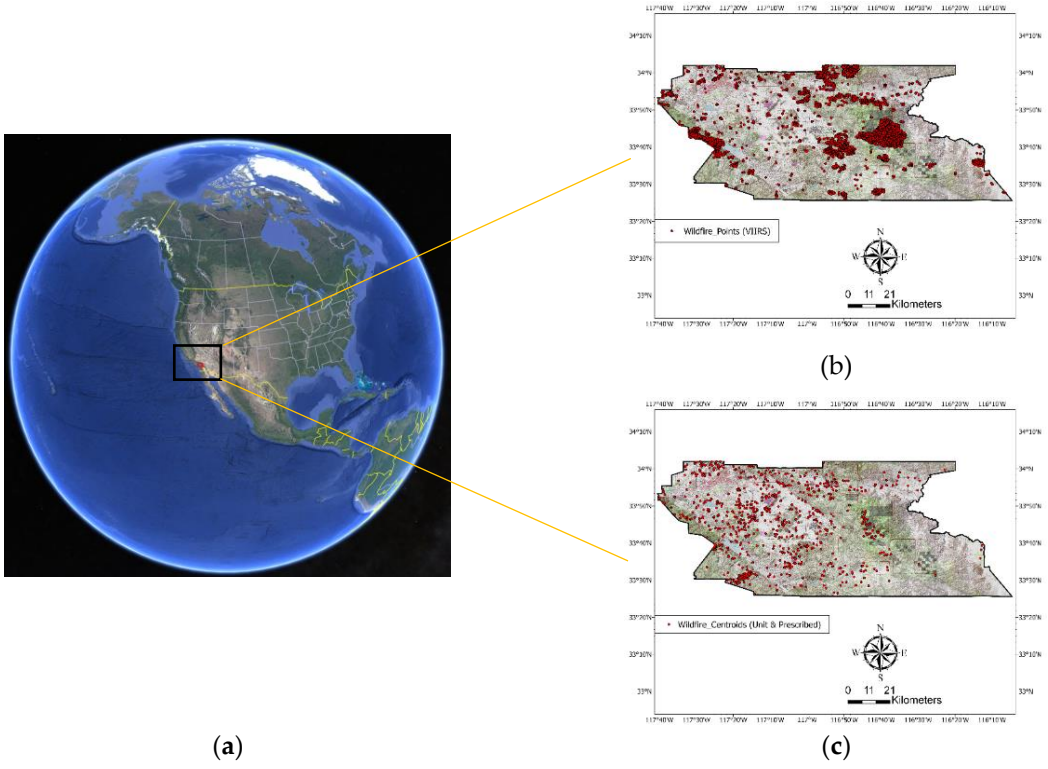



**Figure 1.** Area of study: (a) General location in the globe; (b) Area topo map and the wildfire locations in red; (c) Area topo map and the unit and prescribed wildfire centroids in red.

### *2.2. Wildfire Controlling Factors*

Identifying and inputting vital and diverse controlling factors is a crucial requirement for wildfire susceptibility modeling [9]. Thus, four groups of controlling factors—topography, vegetation, spatial accessibility, and climate—comprising a total of 18 factors, were selected for this research. The details of these factors are provided in Table 1. All factors were resampled to a 30 m × 30 m resolution, and the WGS 1984 UTM Zone 11N coordinate system was used.

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**Table 1.** Details of susceptibility controlling factors.

| Group | Controlling Factor | Description | Reference |
|---|---|---|---|
| Topography | Elevation | Affects accessibility and climate variability ([10];[11]) | SRTM |
| | Aspect | Difference in solar radiation hence temperature and moisture [10] | |
| | Slope | Causes vegetation distribution and fire spread speed changes [12] | |
| | Curvature | Influence on rate of spread [13] | |
| | TWI | Controls on soil moisture by topography [14] | |
| | SPI | Represents the water erosive power [15] | |
| Vegetation | NDMI | Reflects the fuel moisture content [16] | Sentinel-2 |
| | NDVI | Exhibits the vegetation status and type [17] | |
| Spatial accessibility | Land use | Impacts on availability and buildup of fuel [18] | Sentinel-2 |
| | Population density | Suggests increase in ignition points [19] | CIESIN |
| | Distance to roads<br>Distance to urban areas | Incorporate the human influence [20] | CENSUS |
| | Distance to rivers | Represents agricultural-linked fuels [21] | USGS |
| Climate | Rainfall | Changes the fuel load and moisture [22] | CEC |
| | Temperature | Affects fuel moisture and heat transfer: radiation and convection [23] | |
| | Solar radiation | Decrease the fuel moisture [24] | NREL |
| | Wind speed | Accelerates the wildfire spread speed [25] | Global Wind Atlas |
| | RH | More RH enhances fuel moisture recovery [26] | NOAA |

Topography includes elevation, aspect, slope, curvature, the Topographic Wetness Index (TWI), and the Stream Power Index (SPI), as shown in Figure 2. Data from the Shuttle Radar Topography Mission (SRTM) was used to acquire these topographic controlling factors. These factors directly influence fire spread speed and affect climate, vegetation distribution, and soil moisture [10]; [12]; [13]. TWI and SPI represent hydrological aspects; specifically, TWI controls soil moisture [5], while SPI represents the erosive power of flowing water [15] . Both indices were calculated using an SRTM-based Digital Elevation Model (DEM).

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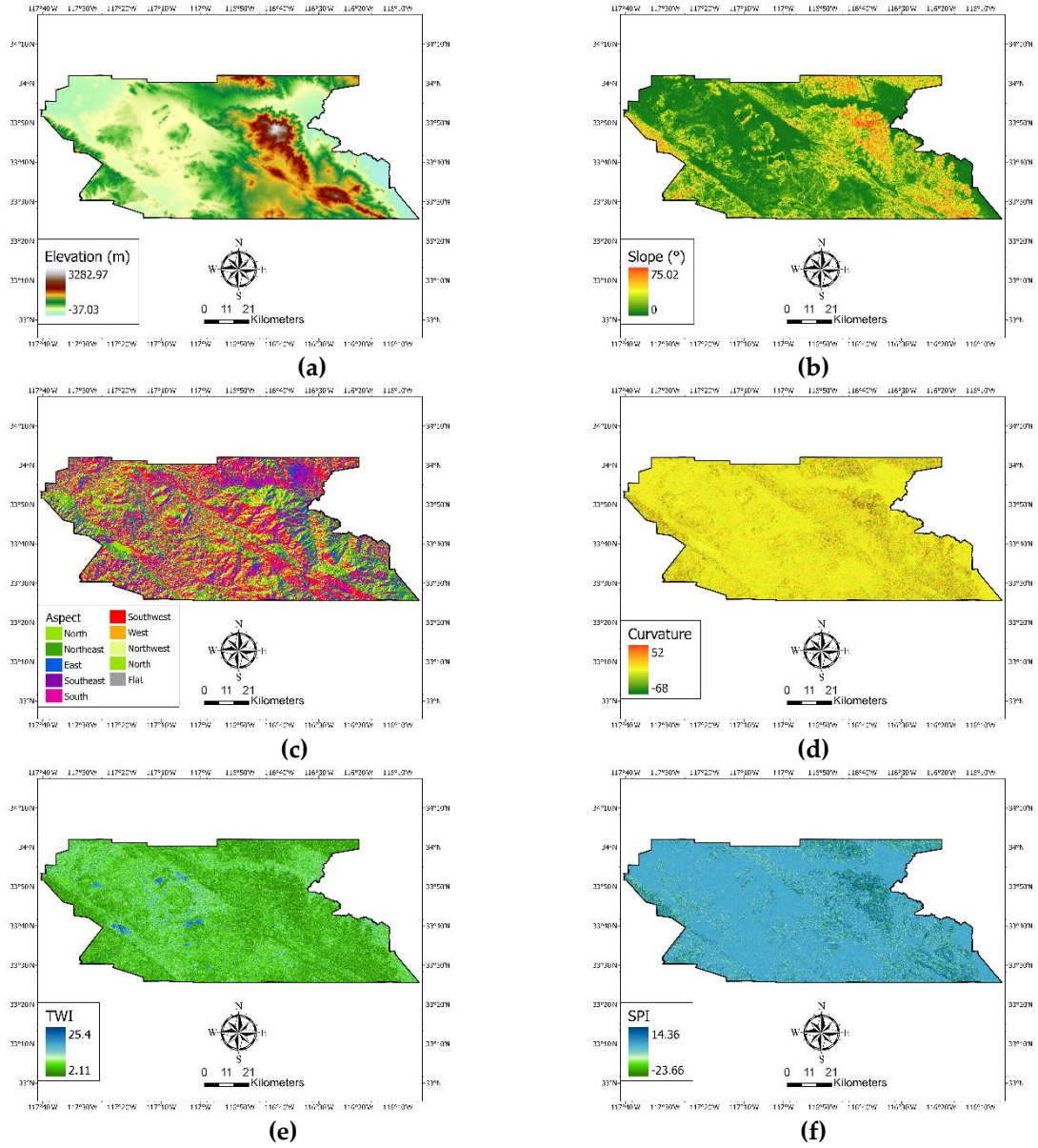



**Figure 2.** Topography controlling factors: (**a**) elevation, (**b**) slope, (**c**) aspect, (**d**) curvature, (**e**) TWI, (**f**) SPI.

The vegetation factors include the Normalized Difference Moisture Index (NDMI) and the Normalized Difference Vegetation Index (NDVI), as illustrated in Figure 3. NDVI values were derived using maximum-value compositing from the peak growing season to minimize atmospheric noise and represent maximum annual vegetation vigor. NDMI was calculated as a multi-year mean of late-summer observations to represent baseline vegetation water content. These factors represent the fuel available for a wildfire: NDVI serves as an index for fuel type [17] , while NDMI provides a measure of fuel moisture [16]. Both factors are derived from Sentinel-2 satellite imagery with a 10 m spatial resolution.

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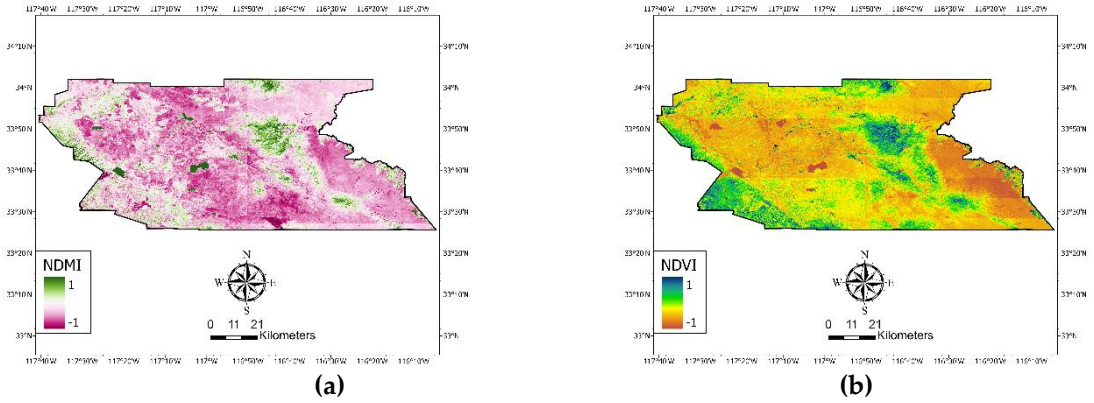
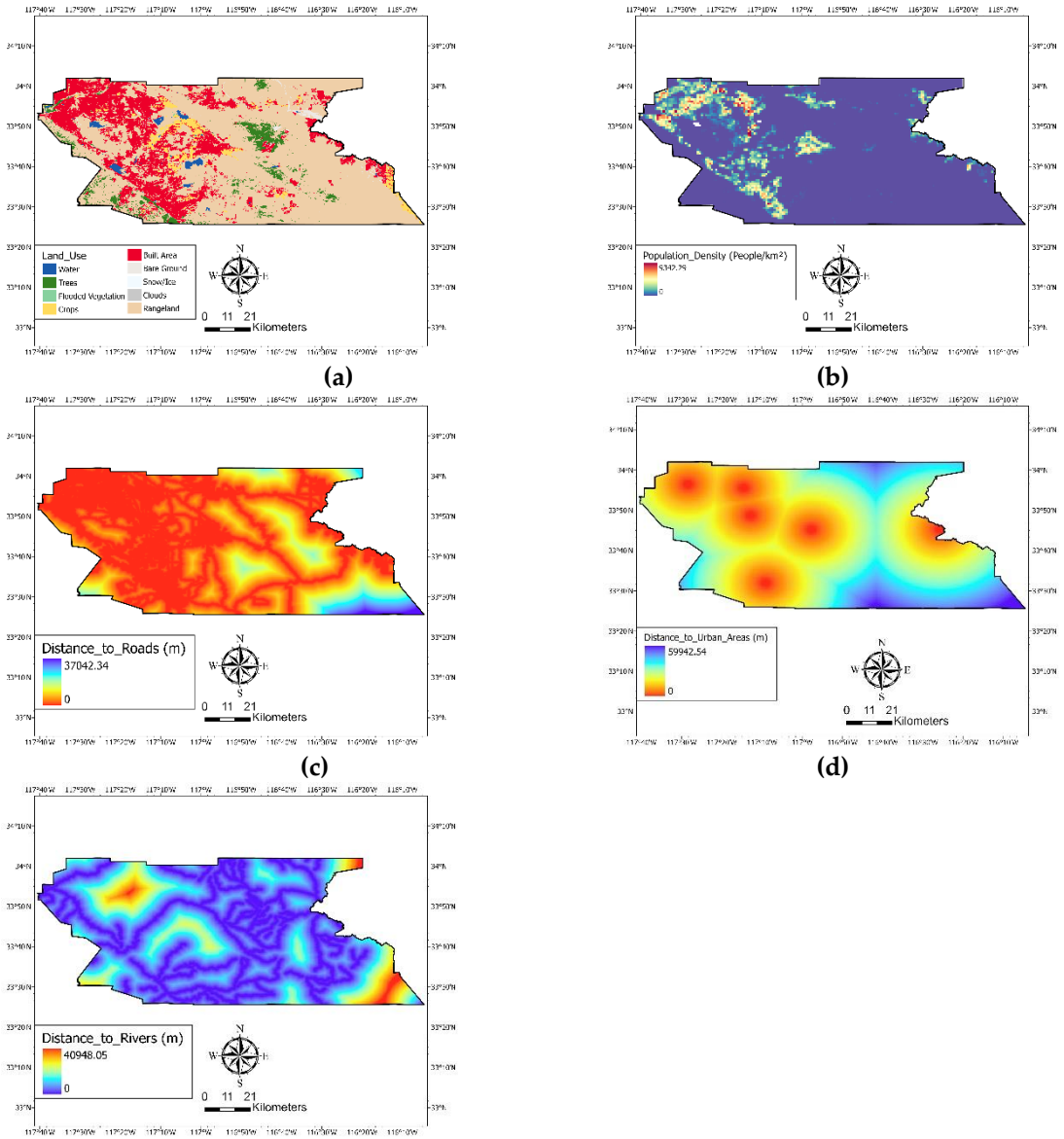



**Figure 3.** Vegetation controlling factors: (**a**) NDMI, (**b**) NDVI.

Spatial accessibility factors include land use, population density, distance to roads, urban areas, and rivers, as depicted in Figure 4. These represent human-related factors that influence wildfire susceptibility. Among these, land use and distance to rivers determine fuel availability, while the remaining three factors affect the frequency of ignition associated with human activities ([18]; [19]]; [20]) . Data for these factors were collected from various sources, namely Sentinel-2, the Center for International Earth Science Information Network (CIESIN), CENSUS, and the USGS.




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**(e)**

**Figure 4.** spatial accessibility controlling factors: (**a**) land use, (**b**) population density, (**c**) distance to roads, (**d**) distance to urban areas, (**e**) distance to rivers.

Figure 5 presents the climate factors, including rainfall, temperature, solar radiation, wind speed, and relative humidity (RH). Rainfall represents long-term multi-year means, calculated by averaging monthly modeled values. Temperature data provides multi-year climatological averages derived from monthly readings. Relative humidity (RH) data were incorporated as multi-year means based on long-term climate normals. Solar radiation data represents multi-year means of the annual average daily total irradiance, designed to account for interannual variability. Wind speed data from the Global Wind Atlas reflects multi-year means, based on multi-year climatological average. Rainfall, temperature, solar radiation, and RH influence fuel load and moisture, while wind speed impacts the wildfire spread rate. Additionally, temperature and solar radiation directly contribute to heat transfer to the fire through radiation or convection ([22]; [23]; [24]; [25]; [26] ). The sources for rainfall and temperature are the Commission for Environmental Cooperation (CEC), while the data for solar radiation, wind speed, and RH are derived from the National Renewable Energy Laboratory (NREL), the Global Wind Atlas, and the NOAA (formerly USGS/NASA) datasets.

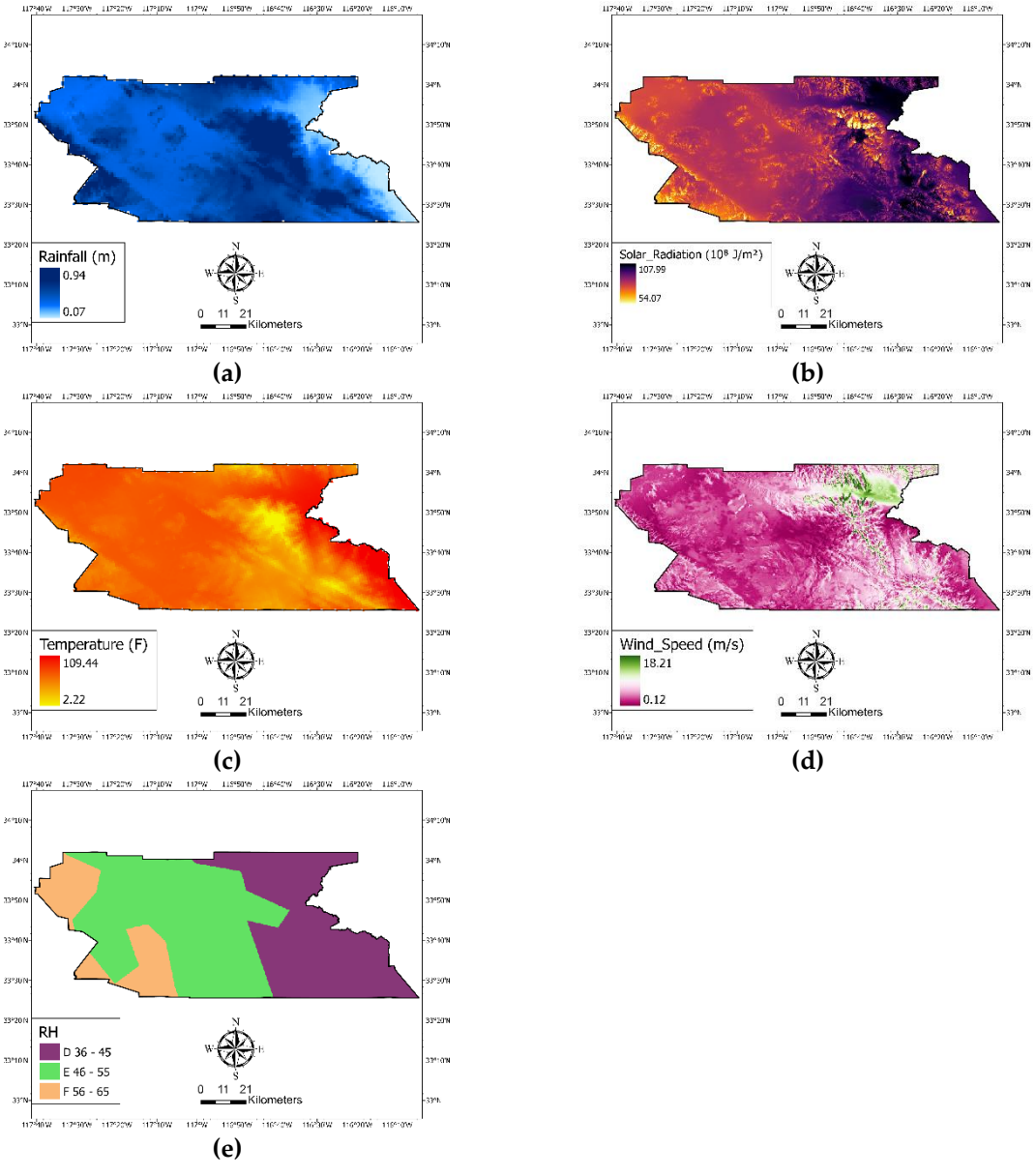



**Figure 5.** Climate controlling factors: (**a**) rainfall, (**b**) solar radiation, (**c**) temperature, (**d**) wind speed, (**e**) RH.

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*2.3. Prediction Setup*

The setup of this study for predicting wildfire susceptibility is shown in Figure 6. In the first stage, 18 potential factors controlling wildfire susceptibility were constructed. In the second stage, the data were preprocessed, and the correlation between the controlling factors was evaluated by means of a multicollinearity test and Pearson correlation. For the multicollinearity test, the variance inflation factor (VIF) was used; a larger VIF represents higher multicollinearity, and a VIF value above 10 suggests that a factor was inappropriately selected due to multicollinearity [27] . Pearson correlation provided a graphical representation of the interrelationship between factors. In the third stage, the data were divided into 70% training and 30% test sets, and ML models—including SVC, LR, ANN, KNN, RF, GB, and LGBM—were established based on the training set. While the 70/30 random split and five-fold cross-validation used in this study provides a standard assessment of model performance, they may not fully account for spatial autocorrelation. This can lead to an overestimation of predictive accuracy, as training and test sets may contain spatially proximal observations that are not truly independent. Future research should implement spatial-block validation strategies to better evaluate model transferability across geographically distinct areas and ensure more conservative, realistic estimates of susceptibility. A five-fold cross-validation method was implemented to determine the optimized model parameters. In the fourth stage, the trained models were evaluated using pixel-based prediction analysis and several ML metrics, including ROC-AUC, accuracy, precision, sensitivity, specificity, F1-score, RMSE, and recall. In this study, a fixed classification threshold of 0.5 was utilized for sensitivity and specificity to determine class membership. Accordingly, the SHAP method was implemented to analyze the contribution of each feature for the highest-performing ML models.

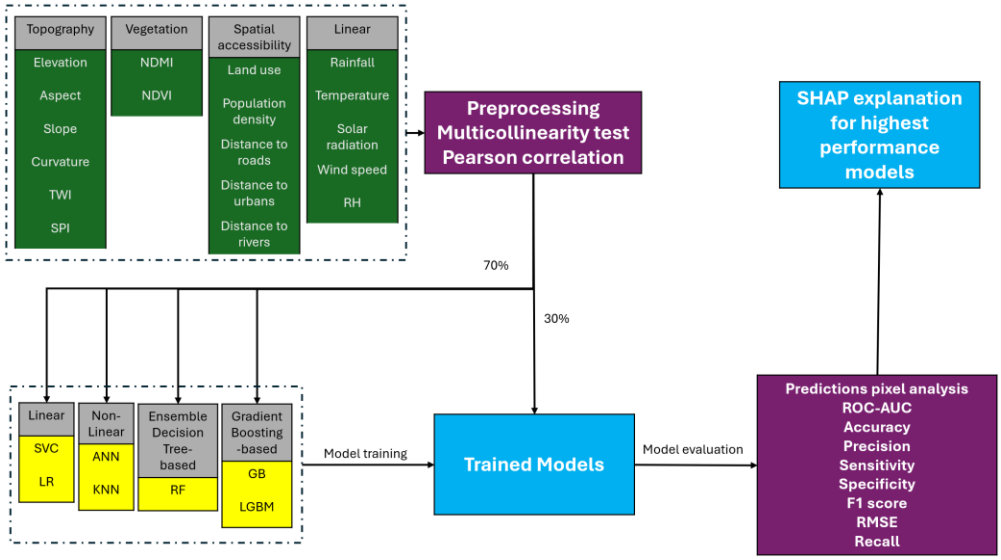



**Figure 6.** wildfire susceptibility ML prediction setup workflow.

The models were tuned using grid search to identify the optimal configuration for the hyperparameter. The optimization focused on minimizing Mean Squared Error (MSE), ensuring the model achieved a balance between predictive accuracy and generalizability. Table 2 shows hyperparameters tuning results for the models.

**Table 2.** Optimal hyperparameter configurations for the evaluated ML models.

| | **Hyperparameters** |
|---|---|
| SVC | C: 0.1, γ: 1, kernel: linear |
| LR | C: 1, solver: lbfgs |
| ANN | batch size: 32, epochs: 100 |

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| | |
|---|---|
| KNN | n_neighbors: 5, metrics: manhatan |
| RF | n_estimators: 300, max_depth: 20 |
| GB | learning rate: 0.5, n_estimators: 200, max_depth: 7 |
| LGBM | num_leaves: 180, learning rate: 0.1, n_estimators: 300 |

## 3. Results

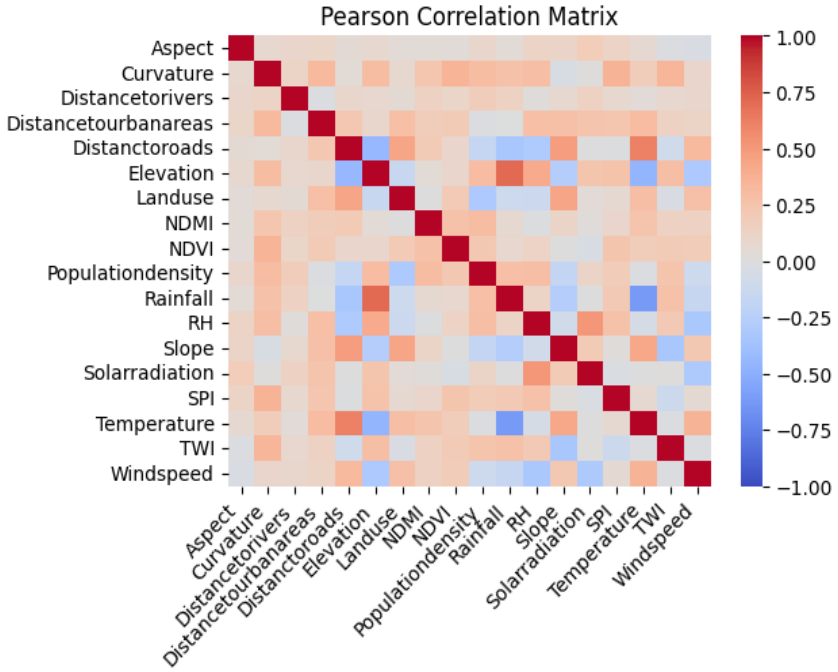
### *3.1. Multicollinearity and Pearson Correlation Assessment*

Table 3 presents the results of the multicollinearity test among the selected wildfire susceptibility factors. The VIF values for all 18 factors are below 4, which is significantly lower than the threshold of 10, indicating that no multicollinearity exists between the wildfire susceptibility factors.

**Table 3.** Multicollinearity assessment results for wildfire controlling factors.

| Factor | VIF | Factor | VIF | Factor | VIF |
|---|---|---|---|---|---|
| Aspect | 1.07 | Land use | 1.71 | Slope | 1.90 |
| Curvature | 1.75 | NDMI | 1.33 | Solar radiation | 1.92 |
| Distance to rivers | 1.15 | NDVI | 1.36 | SPI | 1.61 |
| Distance to urban areas | 1.58 | Population density | 1.64 | Temperature | 3.72 |
| Distance to roads | 2.34 | Rainfall | 3.88 | TWI | 1.71 |
| Elevation | 3.05 | RH | 2.28 | Wind speed | 1.52 |

The Pearson correlation matrix is shown in Figure 7, which was used to determine the correlations between the factors and identify the nature of their relationships. The results show a clear negative or inverse relationship between temperature and rainfall, as well as between temperature and elevation. Conversely, elevation and rainfall, as well as RH and elevation, have a positive or direct relationship with each other.




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**Figure 7.** Pearson correlation assessment matrix results for wildfire controlling factors.

*3.2. Wildfire Spatial Susceptibility Predictions for VIIRS Fires*

In this study, seven ML prediction models are built using the eighteen wildfire susceptibility factors; hence, the wildfire susceptibility values are forecasted for the entire study area. The susceptibility values are divided into five categories including very low, low, moderate, high, and very high; then, they are used to plot wildfire susceptibility maps using quantile symbology in ArcGIS. The wildfire susceptibility prediction maps based on the linear models are presented in Figure 8. Both linear models, the SVC (Figure 8a) and LR (Figure 8b), visually show overall similarities in spatial patterns; however, their specific details are quite different.

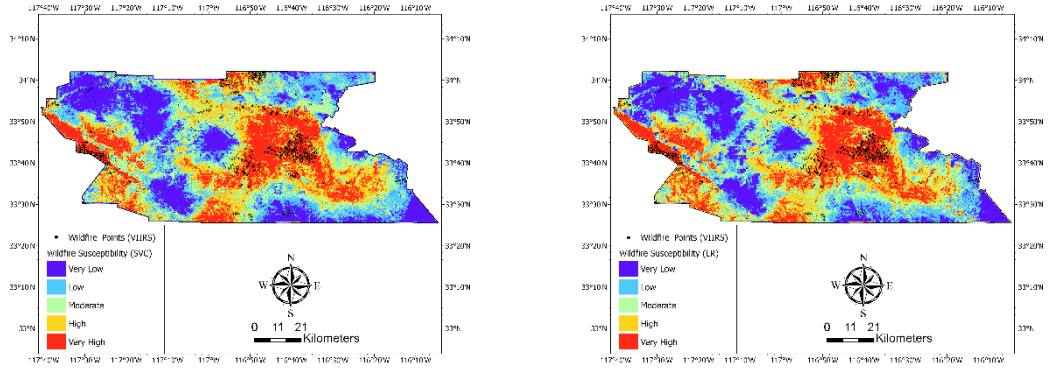



**Figure 8.** Wildfire susceptibility prediction maps output from linear models: (a) SVC (b) LR.

Figures 9a and 9b show the wildfire susceptibility prediction maps for the two non-linear models, the ANN and KNN, respectively. Compared to the linear models, the very high susceptibility regions for these non-linear models are visually smaller in spatial extent. Additionally, the KNN model appears to have difficulty distinctly separating susceptibility categories, especially within the very high susceptibility region.

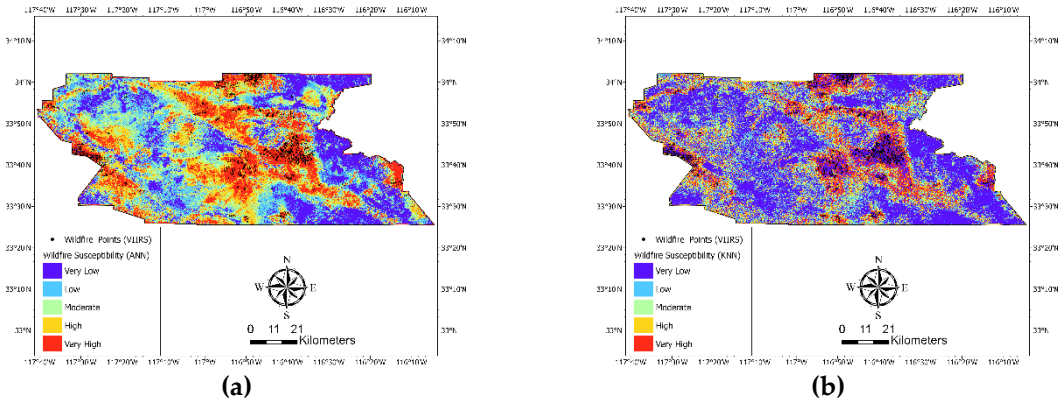



**Figure 9.** Wildfire susceptibility prediction maps output from non-linear models: (a) ANN (b) KNN.

The wildfire susceptibility prediction map for the RF model is shown in Figure 10. As the only ensemble decision tree-based model used in this study, the RF model visually provides a slightly clearer distinction among the susceptibility categories compared to the non-linear models.

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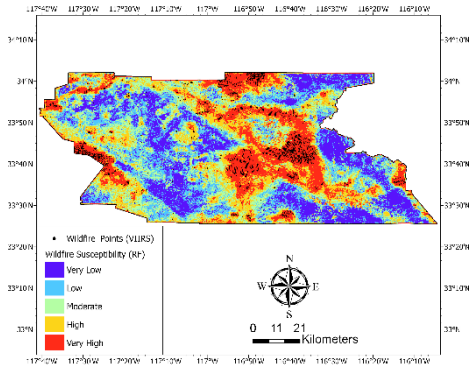
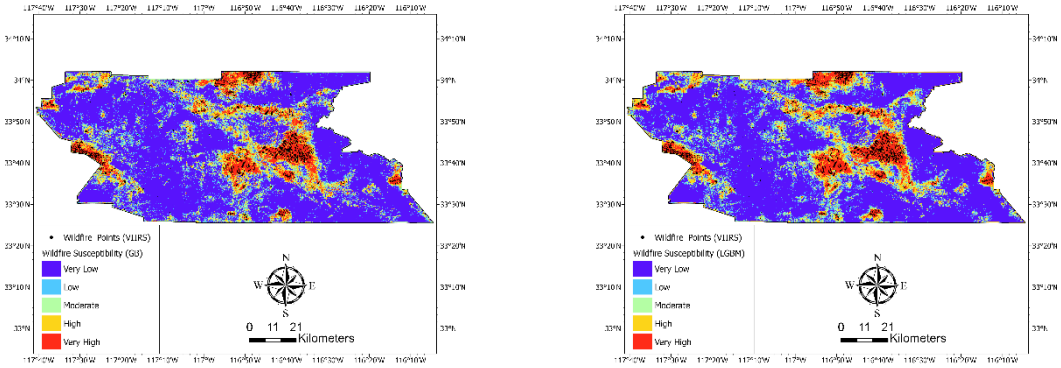



**Figure 10.** Wildfire susceptibility prediction map output from the ensemble decision tree-based model: RF.

Figure 11 shows the wildfire susceptibility prediction maps for the gradient boosting-based models, which represent smaller spatial extents for higher susceptibility regions compared to the three classes of ML models previously discussed. Additionally, a comparatively higher number of wildfire data points fall within the very high susceptibility areas, indicating superior prediction performance for both the GB (Figure 11a) and LGBM (Figure 11b) models. Across all four classes of ML models, very high susceptibility is concentrated in eastern areas, with smaller clusters also present in the north and west.




**(a)** **(b)**

**Figure 11.** Wildfire susceptibility prediction maps output from gradient boosting-based models: (**a**) GB (**b**) LGBM.

### *3.3. Predictions Performance Evaluation for VIIRS Fires*

In order to compare the performance of the ML models in predicting the susceptibility categories, the percentage of fire data points and the percentage of map pixels within each wildfire susceptibility category are displayed in Figure 12a and Figure 12b, respectively. From Figure 12b, the very high susceptibility areas for the KNN, GB, and LGBM are smaller than those of the other ML models; however, Figure 12a shows that the KNN model has a much smaller number of wildfire data points in this category. Compared to all other models, the gradient boosting-based GB and LGBM models have the highest number of wildfire data points in the very high susceptibility areas, with the LGBM performing slightly higher than the GB. By capturing the largest number of wildfire data points within the smallest spatial extent of the 'very high' category, the gradient boosting-based models are the most effective at classifying wildfire susceptibility.

The ratio of the fire percentage to the pixel percentage is plotted in Figure 12c to depict the categorization power of the ML models. A higher ratio indicates that more fire data points fall within a smaller spatial extent of the susceptibility categories, revealing better categorization ability for the ML model. For the majority of the ML models, this ratio increases for higher susceptibility categories, which relates to the sensitivity of the susceptibility classification. With the exception of the KNN

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model, which shows a low ratio below 1, all other models have a ratio above 3; the ratios for the gradient boosting-based models are significantly higher, exceeding 13.

While the wildfire susceptibility pixel analysis described in Figure 12a–c is influenced by the thresholds of the susceptibility categories, the ROC curve shown in Figure 12d is independent of these thresholds and serves as a robust metric for the ML models' performance evaluation. In other words, feature values were categorized into quantiles to mitigate the influence of outliers and linearize non-monotonic associations, while ROC-AUC was utilized as a threshold-independent complement to evaluate the model's discriminative power across all possible decision boundaries. Accordingly, the AUC values for the LR and SVC models are the lowest, at approximately 0.8, indicating the poorest predictive performance. Conversely, the RF, GB, and LGBM models have AUCs above 0.95, with the LGBM achieving the highest AUC of 0.971, which translates to the superior predictive performance among the models.

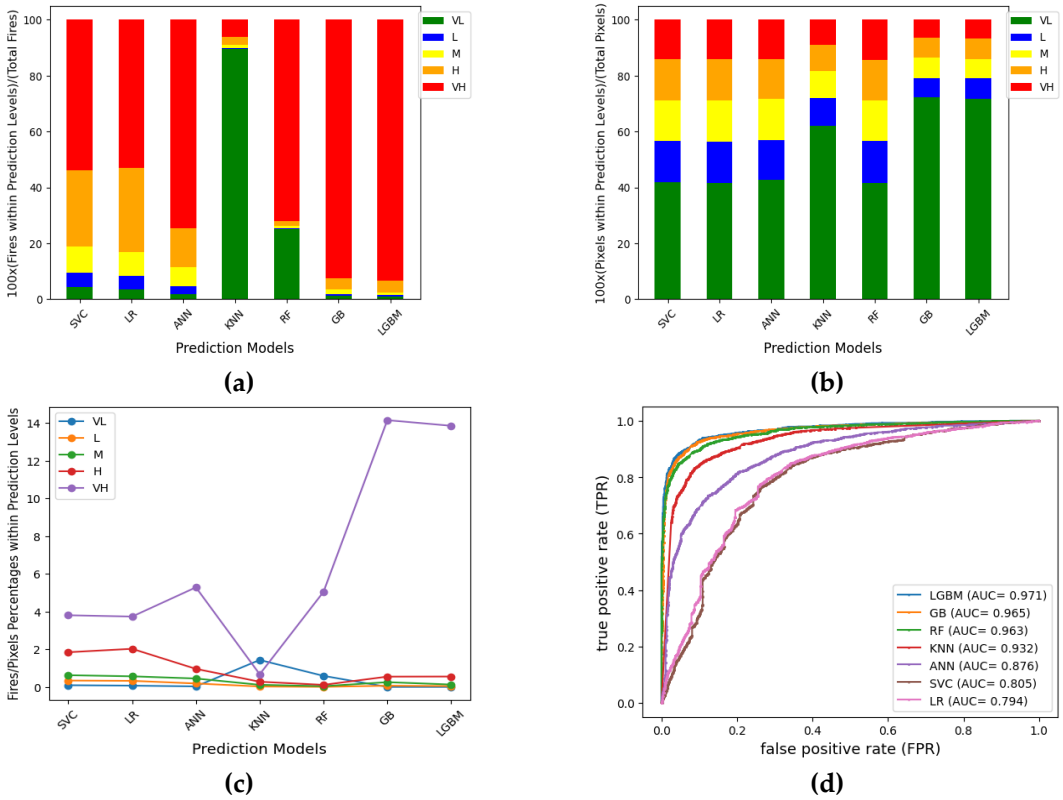



**Figure 12.** Maps analysis results and ROC: (**a**) percentage of fire data points per categories, (**b**) percentage of pixels within categories, (**c**) fire percentage divided by pixel percentage and (**d**) ROC.

The detailed prediction metrics for the ML models' performance evaluation are presented in Table 4. The RF, GB, and LGBM models show remarkable prediction accuracy, precision, sensitivity, specificity, and F1-scores of approximately 90% and above, as opposed to the other models, which have relatively lower values for these metrics. Besides, the RMSEs for these models show a relatively smaller predictive error. Across all metrics except for precision, the LGBM exhibits the highest performance.

**Table 4.** ML models performance metrics.

| | Accuracy (%) | Precision (%) | Sensitivity (%) | Specificity (%) | F1 score (%) | RMSE |
|---|---|---|---|---|---|---|
| SVC | 74.08 | 68.94 | 84.98 | 63.78 | 76.12 | 0.51 |
| LR | 75.10 | 71.40 | 82.25 | 68.21 | 76.44 | 0.50 |
| ANN | 80.45 | 85.62 | 71.84 | 88.59 | 78.13 | 0.44 |

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| | | | | | | |
|---|---|---|---|---|---|---|
| KNN | 86.34 | 84.75 | 87.67 | 85.07 | 86.18 | 0.37 |
| RF | 89.97 | 90.03 | 89.50 | 90.43 | 89.76 | 0.32 |
| GB | 89.97 | 91.60 | 91.01 | 92.10 | 91.30 | 0.32 |
| LGBM | 91.65 | 91.39 | 91.63 | 91.67 | 91.51 | 0.29 |

Table 5 shows the summary of recent similar studies spanning from 1996 to 2023, detailing their best performance ML model approach plus information about their inventory data, number of factors, and best performance metrics. Among all model approaches including Random Forest (RF), Decision Trees (DT), and Deep Learning (DL) the Gradient Boosting (GB) based model used in this study is the only model that leads to high model performance with a high number of inventory fire data of more than 1000 points. Specific inventory records indicate that this study utilized 6,366 active wildfire points across 18 features, yielding a significantly lower model variance and an RMSE of 0.29. While [28] reported a slightly higher random-split AUC (98%) using 510 historical fire points in a Mediterranean climate, direct comparisons across distinct study areas and sample boundaries cannot be verified on AUC values alone. Future comparisons would require normalized performance curves across identical spatial dataset scales.

**Table 5.** ML models performance comparison with other existing studies.

| | Best Performance model approach | Years | Wildfire Points | Factors | Best performance metrics |
|---|---|---|---|---|---|
| Tonini et al. | RF based | 1997-2017 | 365 | 9 | RMSE = 0.50 |
| Bountzouklis et al. | RF | 1997-2020 | 48038 | 36 | F1-score = 0.87 |
| Singha et al. | RF based | 2012-2022 | 500 | 44 | AUC = 89% |
| Phelps and Woolford | DT | 1996-2016 | 550 | 11 | AUC = 90% |
| Abdollahi and Pradhan | DL | 2019-2020 | 521 | 11 | AUC = 93% |
| Yue et al. | GB based | 2013-2022 | 12462 | 17 | AUC = 93% |
| Gholamnia et al. | RF | 2012-2017 | 34 | 16 | AUC = 94% |
| Mohajane et al. | RF based | 2005-2014 | 510 | 10 | AUC = 98% |
| This study | GB based | 2012-2023 | 6366 | 18 | AUC = 97%<br>RMSE = 0.29 |

### *3.4. Highest Performance Model Further Analysis*

#### 3.4.1. TreeSHAP Results for VIIRS Fires

Given the superior classification performance of the gradient-boosted tree framework, the TreeSHAP variant [29] was implemented to confirm feature attribution consistency across the top-performing ensemble tree classes. Figure 13a displays the absolute SHAP values of the factors on the X-axis and the factors on the Y-axis, ranked by their importance. Therefore, the first factor at the top makes the greatest contribution, while the last factor at the bottom makes the least contribution to determining wildfire susceptibility. Accordingly, rainfall, solar radiation, distance to urban areas, elevation, temperature, distance to roads, distance to rivers, NDVI, and RH have the most significant impact on wildfire susceptibility.

In order to visualize the effects of the factors alongside their importance, the directional SHAP values in a beeswarm plot are shown in Figure 13b. In this figure, the actual SHAP values are provided, where a positive value indicates a positive impact and a negative value indicates a negative impact on wildfire susceptibility; high and low feature values are represented by red and blue, respectively. Overlapping points are randomly shifted along the Y-axis to enhance the visualization of the SHAP value distribution for each feature and avoid distortion. Certain features, including solar

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radiation, distance to urban areas, land use, and temperature, show an overall positive trend in their impact on wildfire susceptibility. On the other hand, features such as rainfall, RH, elevation, TWI, and population density reveal a negative trend. Both sets of factors exhibit some variability in their trends, as the data points do not show a uniformly low or high feature value corresponding strictly to high or low SHAP values.

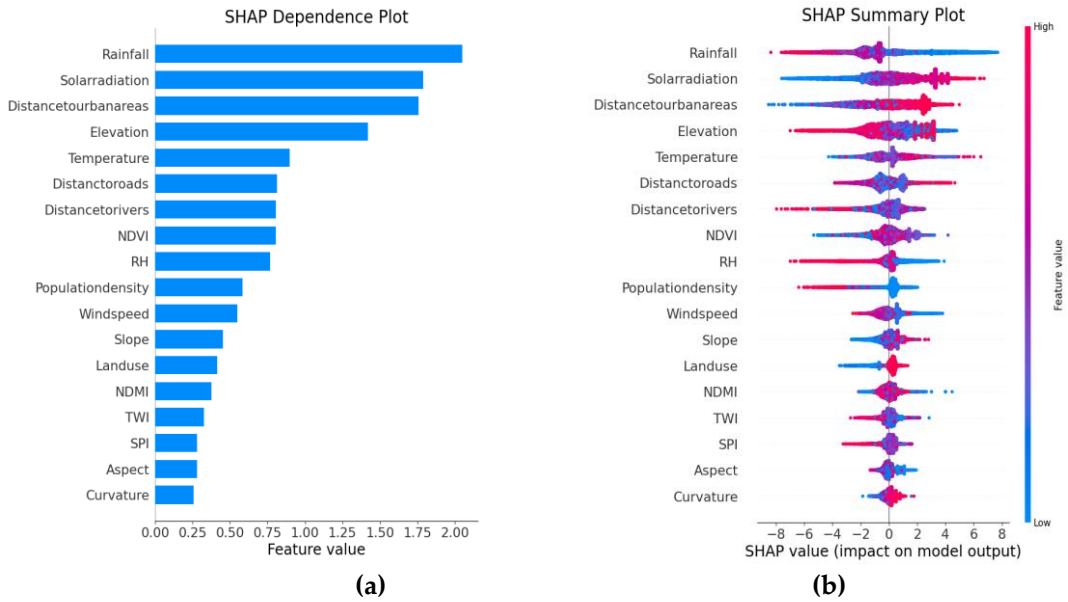



**Figure 13.** Factors importance visualization: (a) dependence plot (b) summary plot.

Figure 14 shows the individual impact of the four most influential factors on SHAP values. These factors are plotted on the horizontal axes, while the SHAP values are shown on the vertical axes. The SHAP value trend versus rainfall, shown in Figure 14a, illustrates the negative impact of rainfall on wildfire susceptibility. Depending on the timing, summer rainfall from thunderstorms increases fuel moisture, which diminishes fuel flammability. Conversely, rainfall received during 'normal' winter months can enhance wildfire occurrence by providing ample grass fuel loadings. The slope of wildfire susceptibility decreases sharply between 200–400 mm, remains relatively steady up to 600 mm, and drops sharply at approximately 630 mm. In contrast, solar radiation (Figure 14b) has a positive influence on wildfire susceptibility, increasing up to $90 \times 10^8$ J/m$^2$ before remaining nearly steady; this is due to the decrease in fuel moisture associated with increased solar radiation. Although distance to urban areas (Figure 14c) has an overall positive effect on wildfire susceptibility, it shows a slight negative trend in close proximity to urban areas (below 36 km). Wildfire fuels are generally distant from urban centers; however, at closer distances, susceptibility may increase due to human-related activities. Furthermore, the effect of distance from urban areas might be related to proximity to fire suppression resources located in towns and cities. Elevation (Figure 14d) generally has a negative impact on wildfire susceptibility, particularly between 1,250 m and 3,000 m. As elevation increases, climatic conditions shift toward colder temperatures and higher rainfall, resulting in lower fire proneness.

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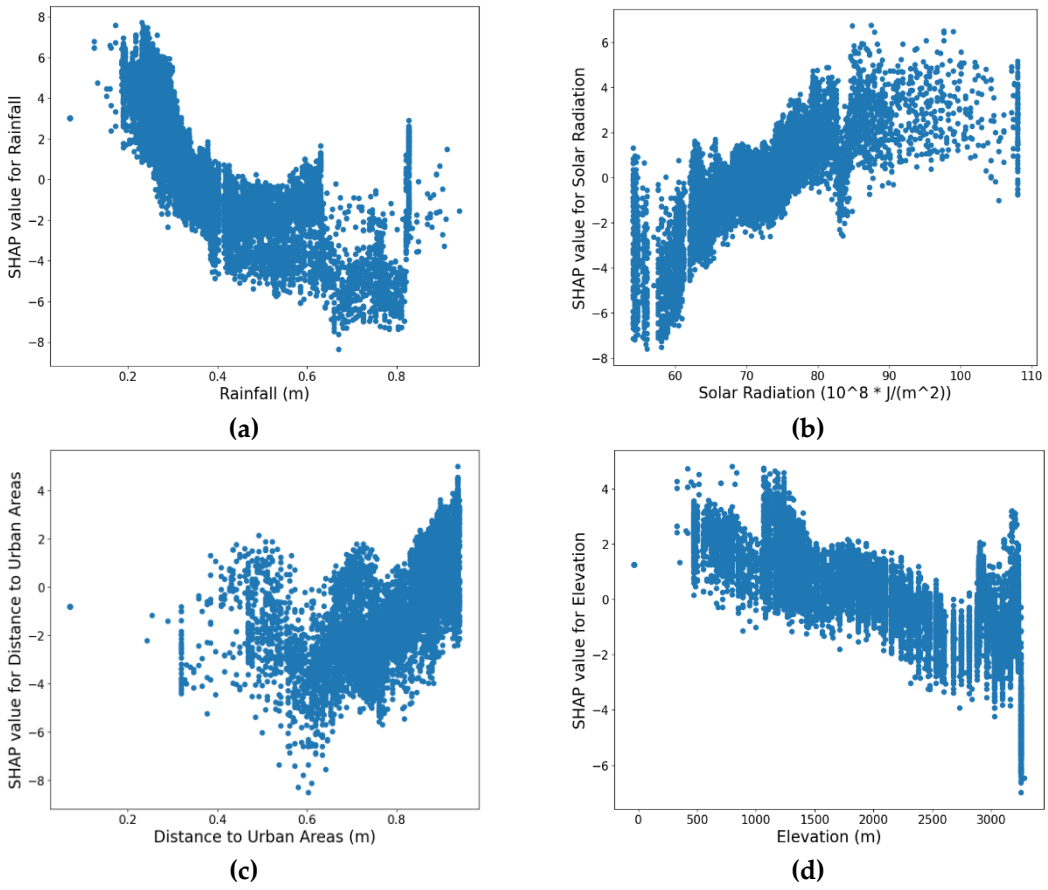



**Figure 14.** Key factors SHAP value: (a) rainfall (b) solar radiation (c) distance to urban areas (d) elevation.

3.4.2. Confusion Matrix, Success and Prediction Rates for VIIRS Fires

Figure 15 further evaluates the performance of the LGBM model in identifying VIIRS fires through two distinct lenses: classification accuracy and spatial predictive power. Figure 15(a) presents the confusion matrix, which demonstrates high classification reliability; the model correctly identified 1,719 fire instances and 1,782 non-fire instances, maintaining a balanced error rate with only 157 false negatives and 162 false positives. Figure 15(b) illustrates the success and prediction rate curves, which measure how efficiently the model performs in predicting high-susceptibility areas. The success rate (AUC = 0.748) indicates a strong fit for the training data, while the prediction rate (AUC = 0.740) confirms the model's capability to generalize to unseen data. Because the gap between the two AUC values is minimal (0.008), the results suggest that the model is well-generalized and free from significant overfitting, making it a reliable tool for fire susceptibility mapping.

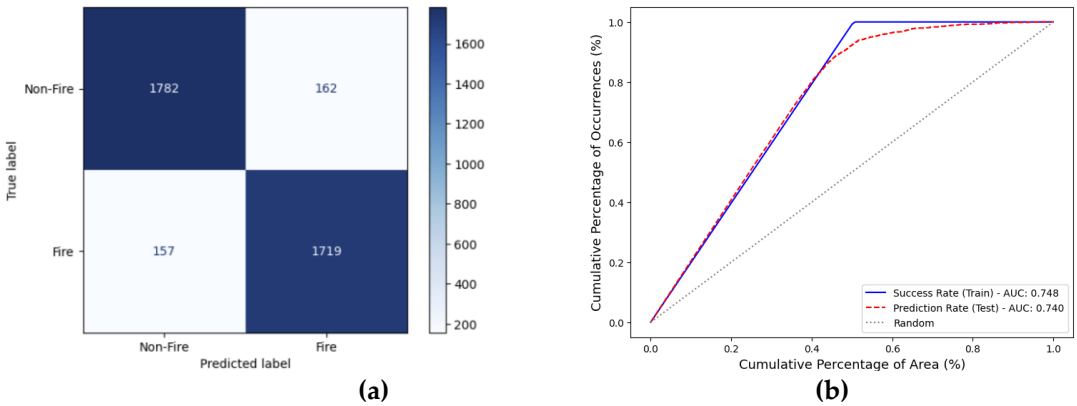



**Figure 15.** Performance metrics for VIIRS fires: (**a**) confusion matrix (**b**) success and prediction rates.

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3.4.3. Spatial Susceptibility Predictions for Unit and Prescribed Fires

To evaluate model performance on controlled ignition scenarios, a separate experiment was conducted by retraining the model on a "unit + prescribed fires" dataset mentioned in section 2.1. While utilizing similar labels to the primary VIIRS wildfire data, this distinct stream incorporated additional variables and fire events missing from the original source. This approach allowed for a comparative analysis of predictive accuracy across different fire types while maintaining the integrity of the initial training set.

Figure 16 shows the map for the LGBM-based wildfire susceptibility prediction for the unit and prescribed fires. The model predicts the majority of the wildfires correctly within the high wildfire susceptibility zone, although some wildfires in the eastern part of the central area incorrectly fall within low susceptibility zones. This might relate to a geographical exception: further east into Riverside County, beyond Cabazon and Pinyon Pines, wildfire occurrence is dependent on the above-normal rainfall received during certain winter months. Increased rainfall leads to higher fuel loading, especially grass fuels, which are key to active fire seasons in this area. Years where Palm Springs receives 200% of normal rainfall result in significant wildfire years, for example, 2013. Similar observations of high wildfire susceptibility related to seasons with higher overall precipitation have been reported for the state of California [30] . However, consistent with the overall data points, most high wildfire susceptibility zones are located toward the western side of the study area.

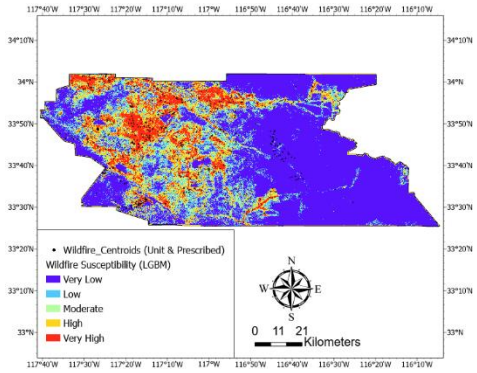



**Figure 16.** Wildfire susceptibility prediction map output from LGBM.

3.4.4. Confusion Matrix, Success and Prediction Rates for Unit and Prescribed Fires

Table 6 summarizes the LGBM model's performance for the unit and wildland fires. The model demonstrates high classification power, with accuracy and F1-scores above 90%. Comparing the 91.57% accuracy to results from the VIIRS fire model, which had a slightly higher accuracy of 91.65%, this retrained model shows a wider gap between sensitivity and specificity. This reflects the increased complexity and specialized nature of the unit and wildland fire data compared to the VIIRS dataset.

**Table 6.** LGBM model performance metrics.

| | Accuracy (%) | Precision (%) | Sensitivity (%) | Specificity (%) | F1 score (%) | RMSE |
|---|---|---|---|---|---|---|
| LGBM | 91.57 | 90.62 | 93.09 | 90.10 | 91.84 | 0.29 |

Figure 17 further illustrates the performance of the retrained model specifically for unit and prescribed wildland fires. Figure 17(a) shows a robust confusion matrix with 512 correctly identified fire instances and 477 correctly identified non-fire instances. The model maintains high precision with relatively few errors, specifically 53 false positives and 38 false negatives. Figure 17(b) displays the success and prediction rate curves, where the model achieved a success rate (AUC = 0.752) and a prediction rate (AUC = 0.732). Although this model shows a larger gap in susceptibility metrics

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compared to the VIIRS-based model, it remains both effective and highly stable when generalizing to unseen data.

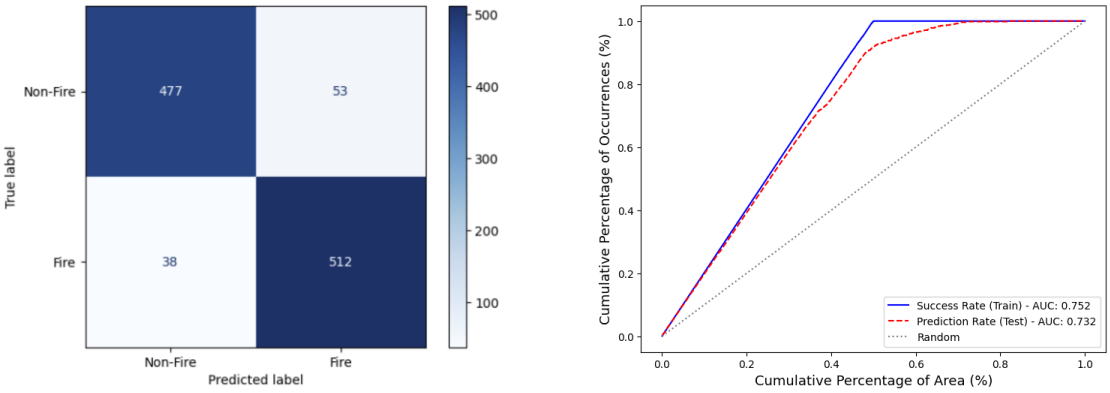



**Figure 17.** Performance metrics for unit and prescribed fires: (**a**) confusion matrix (**b**) success and prediction rates.

## 4. Discussion

As wildfire occurrences and complexities increase, ML models have enhanced the quality of wildfire susceptibility mapping. By integrating remotely sensed data with diverse natural and anthropogenic covariates, this study aligns with the multi-source data fusion approach proposed by [3] , which enhances the interpretability and planning relevance of risk assessments across complex ecosystems. These susceptibility outputs provide a critical foundation for the holistic, performance-based wildfire engineering framework described by [2] , allowing for the prioritization of mitigation zones and preparedness investments within the wildland–urban interface (WUI). While the machine learning models identify strong associations between factors such as solar radiation or rainfall and wildfire occurrence, it is important to note that these importance rankings indicate predictive contribution rather than direct causal mechanisms. Nevertheless, situating these high-susceptibility mappings within a broader resilience toolkit enables more targeted interventions where exposure and community access intersect. This study evaluated the performance of various ML algorithms the SVC, LR, ANN, KNN, RF, GB, and LGBM in producing susceptibility maps for an area in Riverside County. Among all models, the performance of the gradient boosting-based GB and LGBM outperformed the others across multiple metrics. These results are congruent with other studies ([31]; [32]) that have demonstrated enhanced results in predicting wildfire susceptibility.

The most impactful wildfire controlling factors were determined by means of the SHAP technique, with rainfall, solar radiation, distance to urban areas, and elevation at the top of the list. Surface fuel moisture was found to be highly affected by rainfall, indicating an increase in wildfire susceptibility during periods of lower rainfall [11] . Increased rainfall can also elevate fuel and soil moisture, thereby diminishing wildfire susceptibility [33] . The impact of solar radiation on wildfire initiation and spread was reported to be substantial [34] due to resulting elevated temperatures, dried fuel, and lower RH [35]. Some studies have observed that distance to urban areas impacts wildfire susceptibility notably in a positive way, due to the correlation between wildfire occurrence and human activity [36]. On the other hand, the impact was negative at distances very close to urban areas due to negligible or no wildfire fuel availability ([37]; [38] ). Elevation also revealed powerful effects on wildfire susceptibility by influencing wildfire fuel availability and distribution.

Implementing the most appropriate wildfire susceptibility modeling setup the LGBM for the selected study area followed by an explanation using the SHAP technique, will enhance decision support systems in comprehending both model results and controlling factors; hence, it will improve the management of wildfire hazards. The current geographical limitation of this study can be addressed in future work by analyzing a wider region. However, such an expansion may increase

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the complexity of the analysis due to the nature of the historical data and its variability across a larger spatial extent.

## 5. Conclusions

Spatial assessment of wildfire susceptibility is a critical component of emergency management, enabling planning and recovery efforts to diminish wildfire-related risks to people, property, and lives. This study developed a framework to evaluate wildfire susceptibility by integrating eighteen controlling factors in an area of Riverside County and provided susceptibility maps. Among the seven ML models examined, the LGBM revealed extraordinary results in both ML performance metrics and wildfire susceptibility categorization. Thus, the SHAP method was implemented to interpret the model results and generate feature importance. The SHAP outcome indicated that certain controlling factors, including rainfall, solar radiation, distance to urban areas, and elevation, had the most significant impact on wildfire susceptibility predictions. The outcomes of this research underscore the wildfire susceptibility in the Riverside area and facilitate decision-making to execute strategic plans to protect the region against wildfires.

**Acknowledgments:** The views of this paper are my own and do not necessarily reflect those of my employer.

## Abbreviations

The following abbreviations are used in this manuscript:

| | |
|---|---|
| ML | Machine Learning |
| DL | Deep Learning |
| RMSE | Root Mean Squared Error |
| ANN | Artificial Neural Network |
| SVM | Support Vector Machine |
| NDVI | Normalized Difference Vegetation Index |
| NDMI | Normalized Difference Moisture Index |
| GIS | Geographic Information Systems |
| VIIRS | Visible Infrared Imaging Radiometer Suite |
| FIRMS | Fire Information for Resource Management System |
| DEM | Digital Elevation Model |
| SHAP | SHapley Additive exPlanations |
| SVC | support vector classification |
| LR | logistic regression |
| ANN | artificial neural network |
| KNN | K-nearest neighbors |
| RF | random forest |
| GB | gradient boosting |
| LGBM | Light Gradient Boosting Machine |
| AUC | Area Under the Curve |
| AHP | analytical hierarchy process |
| ROC | receiver operating character |
| FWI | Fire Weather Index |
| DT | Decision Tree |
| RF-GBS | Gravitational Based Search |
| RF-PSO | Particle Swarm Optimization |
| SRTM | Shuttle Radar Topography Mission |
| CIESIN | Center for International Earth Science Information Network |
| USGS | United States Geological Survey |
| NOAA | National Oceanic and Atmospheric Administration |
| NREL | National Renewable Energy Laboratory |
| CEC | Commission for Environmental Cooperation |
| TWI | topographic wetness index |
| SPI | stream power index |
| RH | relative humidity |

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VIF variance inflation factor
XGBM extreme gradient boosting machine

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